

PATARA TRIRAT

AI Research Scientist KAIST
Ph.D.

Affiliation: Principal AI Research Scientist at DeepAuto.ai

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RESEARCH INTERESTS

As an AI research scientist, I am passionate about advancing agent-based task automation toward human-agent collaboration and exploring new frontiers of perception in autonomous agents operating in complex, real-world environments through multimodal large language models. My research interests include

Agentic AI & LLM Agents: TS-Debate [[arXiv](#)], AutoML-Agent [[Paper](#)], Agentic Predictor [[Paper](#)], MONAQ [[Paper](#)], ZeroPOIRec [[Paper](#)], OmniRetrieval [[arXiv](#)]

Spatiotemporal Data Mining: DF-TAR [[Paper](#)], MG-TAR [[Paper](#)], Universal Time Series Survey [[arXiv](#)], Time-PEFT [[Paper](#)], MacroLens [[arXiv](#)]

AutoML: AutoML-Agent [[Paper](#)], Conversational AutoML [[Paper](#)], PASTA [[Paper](#)], TFAS [[Paper](#)]

EDUCATION

AUGUST 2020 — FEBRUARY 2025 / DAEJEON, SOUTH KOREA

KAIST

Ph.D in **School of Computing**, GPA: 3.99 / 4.30

Dissertation: Neural Architecture Search for Time-Series Analysis with Context-Specific Performance Estimators

Advisor: Prof. Jae-Gil Lee

Undergraduate Student Research Mentoring:

- + Phatarapran Saraluck
- + Wirittipol Supdateyarnnakorn

Additional Activity:

- + **KAIST Student Ambassador:** KAIST Office of Admissions
- + **AI Engineer Intern:** DeepAuto Co., Ltd.
- + **Club Member:** Lunatic Street Dance Club
- + **Invited Talks:** KSC2021 Top Conference Session and International Seminar on Policies for Older Drivers

AUGUST 2018 — AUGUST 2020 / DAEJEON, SOUTH KOREA

KAIST

M.S. in **Knowledge Service Engineering** (Currently, Data Science), GPA: 3.80 / 4.30

Thesis: Analysis of Dangerous Driving Behavior from Driving Log Data for Traffic Accident Risk Prediction

Advisor: Prof. Jae-Gil Lee

Additional Activity:

- + **Student Assistant Photographer:** KAIST International Scholar & Student Services (ISSS)
- + **Club Member:** Lunatic Street Dance Club

JUNE 2012 - MAY 2016 / BANGKOK, THAILAND

Kasetsart University

B.S. in **Computer Science with First Class Honors**, GPA: 3.74 / 4.00

Advisor: Prof. Pakaket Wattuya

Additional Activity:

- + **Web Developer** (Co-operative Education): NewKru Co., Ltd. (Supervisor: Prof. Sukumal Kitisin and Prof. Usa Sammapun)
- + **Teaching Assistant:** Web Technology and Web Services (CS443)
- + **Dance Instructor / Club Member:** KU Dance Club

EMPLOYMENT

NOVEMBER 2025 — PRESENT / SEOUL, SOUTH KOREA

Principal AI Research Scientist at DeepAuto Co., Ltd.

JANUARY 2025 — OCTOBER 2025 / SEOUL, SOUTH KOREA

Senior AI Research Scientist at DeepAuto Co., Ltd.

Multi-Agent System Development, Automated Design of Multimodal Agentic AI Systems, Tool Generation, LLM Agents for Time-Series Understanding

JANUARY 2024 — MAY 2024 / SEOUL, SOUTH KOREA

AI Engineer Intern at DeepAuto Co., Ltd.

Automated Machine Learning (AutoML), Large Language Model for AutoML using Multi-Agent Framework

AUGUST 2018 — JANUARY 2025 / DAEJEON, SOUTH KOREA

Research Assistant at Data Mining Lab, KAIST

Spatial-Temporal Data Mining and Analysis, Deep Learning Research, Automated Machine Learning for Time Series and Data Stream

MAY 2017 — AUGUST 2018 / BANGKOK, THAILAND

Frontend Developer at Sobotid Edutainment Co., Ltd.

Built APIs and developed web applications under sobtid.me project (both services for customers and in-house officers) using Node.js on Amazon's AWS and the web application using React.js with Google's Firebase

NOVEMBER 2016 — FEBRUARY 2017 / BANGKOK, THAILAND

Intern at Korean Cultural Center

Advertisement Design, Document Translation, and Event Organization Assistance

JUNE 2015 — NOVEMBER 2015 / BANGKOK, THAILAND

Web Developer at NewKru Co., Ltd.

Researched on MOOC platforms and developed a new MOOC application

PUBLICATIONS

Preprint / Ongoing Work

- + Trirat, P., Kwak J.M., Heo J., Lee H., and Hwang, S.J.
MacroLens: A Multi-Task Benchmark for Contextual Financial Reasoning under Macroeconomic Scenarios
<https://arxiv.org/abs/2606.24950>
- + Trirat, P., Heo J., Lee H., and Hwang, S.J.
Visual Anchoring for Scenario-Guided Forecasting
Ongoing Work
- + Trirat, P., Ha, J., Joo, H., Heo J., Lee H., and Hwang, S.J.
Proactive Problem Solver: LLM Agents for the Enterprise R&D Pipeline
Ongoing Work
- + Trirat, P., Zhang, S., Heo J., and Hwang, S.J.
Agentic Tool Generation with Hierarchical Goal-Driven Co-Planning
Ongoing Work
- + Baek, J., Jeong, S., Park, S., Yeo, W., Kang, M., Trirat, P., Lee, H., Hwang, S.J.
OmniRetrieval: Unified Retrieval across Heterogeneous Knowledge Sources
<https://arxiv.org/abs/2605.29250>

Conference Proceedings

- + Trirat, P., Kwak J.M., Heo J., Lee H., and Hwang, S.J.
Multimodal Collaborative Debate for Zero-Shot Time Series Reasoning
EMNLP 2026 (CORE A*, Main)
- + Na, J., Trirat, P., Park, C., and Lee, J.
Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for

Time-Series Foundation Models
ICML 2026 (CORE A*)

- + Trirat, P., Jeong, W, and Hwang, SJ.
Multi-View Encoders for Performance Prediction in LLM-Based Agentic Workflows
ICLR 2026 (CORE A*), ICML 2025 MAS Workshop (Outstanding Paper)
- + Trirat, P. and Lee, J.
MONAQ: Multi-Objective Neural Architecture Querying for Time-Series Analysis on Resource-Constrained Devices
EMNLP 2025 (CORE A*, Findings)
- + Trirat, P., Jeong, W, and Hwang, SJ.
AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML
ICML 2025 (CORE A*)
- + Nam, Y., Trirat, P., Kim, T., Lee, Y., and Lee, J.
Context-Aware Deep Time-Series Decomposition for Anomaly Detection in Businesses
ECML PKDD 2023 (CORE A, Applied Data Science, Oral Presentation)
- + Na, J., Kim, J. Trirat, P., Park, J., Park, S., Yu, K., and Lee, J.
Interpretable Relation Prediction of Undirected Hypergraphs with Dynamic Bipartite Graphs
KIISE, KCC 2023
- + Trirat, P.*, Nam, Y.*, Kim, T., and Lee, J.
AnoViz: A Visual Inspection Tool of Anomalies in Multivariate Time Series
AAAI 2023 (CORE A*, Demonstration Paper)
- + Trirat, P., Shin, Y., Kim, S., and Kim, M.
Generating a Machine Learning Model with a Few Sentences
KIISE, KSC 2021
- + Trirat, P. and Lee, J.
DF-TAR: A Deep Fusion Network for Citywide Traffic Accident Risk Prediction with Dangerous Driving Behavior
TheWebConf 2021 (CORE A*, Oral Presentation)
- + Trirat, P., Noree, S., and Yi, M.Y.
IntelliMOOC: Intelligent Online Learning Framework for MOOC Platforms
EDM 2020 (CORE B, Poster Paper)
- + Trirat, P., Kim, M., and Lee, J.
Experimental Analysis of the Effect of Autoencoder Architectures on Time Series Representation Learning
KIISE, KCC 2020
- + Shin, Y., Yoon, S., Trirat, P., and Lee, J.
CEP-Wizard: Automatic Deployment of Distributed Complex Event Processing
IEEE ICDE 2019 (CORE A*, Demonstration Paper)

Journal Articles

- + Trirat, P., Shin, Y., Kang, J., Nam, Y., Na, J., Bae, M., Kim, J., Kim, B., and Lee, J.
Universal Time-Series Representation Learning: A Survey
ACM Computing Surveys (2026) Vol. 58 No. 12 (SCIE, Impact Factor: 30.4)
- + Kim, J., Seo, Y., Kim, Y., Kang, J., Shin, J., Trirat, P., and Lee, J.
Large Language Models are Zero-Shot Point-of-Interest Recommenders
Data Mining and Knowledge Discovery (SCIE, Impact Factor: 5.5, ECML

PKDD 2025 Journal Track, Oral Presentation)

- + Trirat, P. and Lee, J.
TFAS: Zero-Shot NAS for General Time-Series Analysis with Time-Frequency Aware Scoring
Machine Learning (SCIE, Impact Factor: 4.9, ECML PKDD 2025 Journal Track, Oral Presentation)
- + Trirat, P. and Lee, J.
PASTA: Neural Architecture Search for Anomaly Detection in Multivariate Time Series
IEEE Transactions on Emerging Topics in Computational Intelligence (SCIE, Impact Factor: 6.5)
- + Trirat, P., Yoon, S., and Lee, J.,
MG-TAR: Multi-view Graph Convolutional Networks for Traffic Accident Risk Prediction
IEEE Transactions on Intelligent Transportation Systems (2023) Vol. 24 No. 4 (SCIE, Impact Factor: 9.1)
- + Trirat, P., Yoon, S., and Lee, J.,
Spatial and Temporal Analysis of Dangerous Driving Behavior Using DTG Data
Korea Database Research (2019) Vol. 35 No. 3 (KCI)
- + Trirat, P., Yoon, S., and Lee, J.,
Study on Driving Behavior of Different Vehicle Types Using DTG Data
Korea Database Research (2019) Vol. 35 No. 2 (KCI)

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ACADEMIC SERVICES

Conference Reviewer

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining
- SIAM International Conference on Data Mining
- International Conference on Machine Learning
- International Conference on Learning Representations
- Annual Conference on Neural Information Processing Systems
- AAAI Conference on Artificial Intelligence
- ICML 2025 Workshop on MAS
- ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving
- ACL Rolling Review (ARR): May 2026

Journal Reviewer

- Transactions on Machine Learning Research
- ACM Computing Surveys
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Mobile Computing
- IEEE Transactions on Emerging Topics in Computational Intelligence
- Journal of Supercomputing

Research Mentoring & Collaboration

- Geonju Lee (Research Intern / Master's Student)
- Jisung Ha (Research Intern)
- Jihye Na (Ph.D. Student)

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LANGUAGES
AND SKILLS

Languages

Thai: Native
English: Effective Operational Proficiency
> **TOEIC:** 975 / 990
Korean: Limited Working Proficiency

- > **TOPIK IBT**: 6 / 6
- > **TOPIK**: 5 / 6
- > **TOPIK Speaking**: 3 / 6

Programming Languages

Python, JavaScript, HTML/CSS, R, Java

Frameworks / Libraries

Tensorflow, Keras, React.js, Node.js, PyTorch

Cloud Services

Google Cloud Platform, Amazon Web Services, Microsoft Azure

Softwares

Microsoft Offices, Apple iWork, Adobe Photoshop, Apple Final Cut Pro

— ACCOMPLISHMENTS

Honors and Awards

- + 2026 *Best Reviewers* (Research Track) by KDD 2026
- + 2026 *2nd Place* of XForecast Challenge at KDD 2026 Finance Day by LG AI Research
- + 2026 *Gold Reviewer Award* (Top 25%) by ICML 2026
- + 2025 *Outstanding Paper Award* by ICML 2025 Workshop on Multi-Agent Systems in the Era of Foundation Models: Opportunities, Challenges and Futures
- + 2025 *Commendable Reviewer* by KDD (February Cycle) 2025
- + 2025 *Top 10% Outstanding Reviewer* by KDD (August Cycle) 2025
- + 2023-24 *Student Ambassador* by KAIST Office of Admissions
- + 2023 *University Visit Program Scholarship Recipient* by KAIST School of Computing
- + 2022 외국인 유학생 독도사랑 한국어 말하기대회 최우수상 (팀명: K-GOOSE) by Daegu University
- + 2020-24 *KAIST Scholarship* (Doctoral Program) by KAIST
- + 2019 3rd Place at *Year End Party Talent Show* by KAIST ISSS
- + 2018-20 *KAIST Scholarship* (Master Program) by KAIST
- + 2017 Winner at *Thailand Hip Hop Dance Championship* (Megacrew Division) by Hip Hop International Thailand
- + 2016 Winner *LINE Finance Night Dance Competition* by LINE Finance
- + 2014-16 *Faculty of Science Awards* for Outstanding Academic Achievement Student by Faculty of Science, Kasetsart University
- + 2013-15 *Medal of Excellent Academic Achievement* by Kasetsart University
- + 2013-15 *5As Scholarship* by Department of Computer Science, Kasetsart University
- + 2012-14 *Tuition fee Exemption* (5As+ per semester) by Kasetsart University
- + 2012 *King Rama IX's Royal Awards* (Qualified Grantee of Big Vocational College) by Ministry of Education of Thailand
- + 2011 3rd Runner Up at *MOS Olympics Competition Thailand* by ARiT
- + 2011 Winner H.R.H Princess Soamsawali Trophy at *SIAM MOS Competition* by ARiT

Certificates

- + 2022.08 Letter of Appointment (*Member of Dokdo Supporters*) by Dokdo Foundation
- + 2019.07 Certificate of Completion (*Introduction to Data Science*) by Skooldio
- + 2019.04 Certificate of Completion (*2019 Spring Mentor Program*) by KAIST ISSS
- + 2018.03 *Mobile Sites Certification* by Google
- + 2018.03 Certificate of Completion (*Modern React with Redux*) by Udemy

- + 2017.08 Certificate of Completion (*React Js and Redux Mastering Web Apps*) by Udemy
- + 2017.05 Certificate of Training (*Mr.Wiggles Hip Hop Camp*) by Studio Zoom
- + 2016.08 Certificate of Achievement (*K-Singing Class*) by Korean Cultural Center
- + 2016.04 *Test of Practical Competency in ICT* (TOPCIT — Score: 297, Level: 2) by IITP and KPC
- + 2015.12 Certificate of Completion (제1기 한국유학대비 한국어과정: 중급) by Korean Education Center
- + 2014.05 Certificate of Participation (*Thailand K-POP Camp for Young Talents*) by Korean Cultural Center and Silpakorn University
- + 2014.01 Certificate of Completion (*5th Master Class*) by Move With Passion
- + 2013.10 Certificate of Honor (*Super Start Summer Camp*) by Monkey Town Academy
- + 2013.08 Certificate of Participation (*ACM-ICPC Thailand Central Group A Programming Contest 2013*) by ACM-ICPC
- + 2013.03 Certificate of Training (*1st Software Developing Skills Training Camp*) by Department of Computer Science
- + 2012.10 Certificate of Training (*1st Programming Skills for Problem-solving Training Camp*) by Depart of Computer Science

Major Course Projects at KAIST

- + Tiny-TimeNAS: A Python Package for Time-Series Analysis on IoT Devices with Tiny Neural Architecture Search (CS565: IoT Data Science)
- + A3M: Adaptive Memory Management Mechanism for Online Task-free Continual Learning (AI602: Advanced Deep Learning)
- + Q&AI: AI-Infused Q&A System for Online Learning (CS492 (J): Special Topics in Computer Science<AI for Smart Life>)
- + Towards Better Understanding of Cryptocurrency: A Multi-Aspect Analysis (CS564: Data Science Methodology)
- + Neural Predictor for Neural Architecture Search in Tensorflow 2.0 (CS470: Introduction to Artificial Intelligence)
- + Mr.TAD: Time Series Anomaly Detection with Multi-resolution Ensemble Learning (CS570: Artificial Intelligence and Machine Learning)
- + Conversational AutoML: A Framework for Generating an Efficient Machine Learning Model with a Few Sentences (CS579: Computational Linguistics)
- + HandyBoard: An Adjustable Split of Soft Keyboard in Landscape Mode Smartphone Touch Screens (CS584: Human-Computer Interaction)
- + Jet Presentation: A Presentation Rehearsal Partner for Everyone (KSE624: Mobile and Pervasive Computing for Knowledge Services)
- + TRepNet: Convolutional Denoising Autoencoder for Time Series Representation Learning (KSE801: Data Analysis based on Deep Learning)
- + Answering Question in Massive Open Online Course by Recommending Ranked Video Clips (KSE643: Knowledge Engineering and Intelligent Decision Making)

Previous Projects

- + Korean Square Web Application (React.js)
- + D Maniac Studio Mobile Application (React Native)
- + D Maniac Studio Web Application (React.js)
- + Sneakavilla Online Store (Wordpress)
- + Kyouei Co., Ltd. Website (Wordpress)

- + Sriracha Tugboat Co., Ltd. Website (Wordpress)
- + Web Application UX/UI Design (Human-Computer Interaction Assignment and Project using AngularJs)
- + Robotz Language and Compiler (Formal Language & Automata Theory Project using Java)
- + Hook Flash Card E-Learning Web Application (Web Technology and Web Service Project#2 using CakePHP)
- + Workpoint Web-based Resource Management (Web Technology and Web Service Project#1 using PHP)
- + Floral Bloom (ERP Project using SAP B1)
- + Flower Bloom Online Store (Ruby on Rails on IBM Bluemix)
- + Adventure in Wonderland (Software Engineering Game Project using Java)
- + Hangman Game (Network Application Project using Python with Tk GUI)
- + Index Sequential Search (Data Structure Project using Adobe Flash Animation)
- + Fan Store (Software Construction Project using Java)
- + Electricity Cost Prediction Program (DevCamp#1 Windows 8 Application Project using C#)
- + Hangman CLI Game (AlgoCamp#1 using Python)
- + Polynomial Calculation (Computer Programming Project using Python)
- + Store Project (Computer Programming Project using Python)

On-stage Performances

- + 2025 CLICK Battle Vol. (Top 4)
- + 2023 Lunatic - Spring Performance
- + 2022.11 Lunatic - Funk Night Vol.4 Tier Battle (Top 2 / **Finalist**)
- + 2022.07 1MDJ Showcase - Best Dancer
- + 2022 Lunatic - Spring 오픈동방 Battle (Top 4)
- + 2019 Lunatic - Spring PPOP Battle (Top 8)
- + 2019 Lunatic - Open Battle (Prelims)
- + 2019 Lunatic - 기획부장배 Battle (Top 8)
- + 2019 Lunatic - Spring 오픈동방 Battle (Guest Show)
- + 2018 Lunatic - Fall Freshmen Battle (Top 8)
- + 2018 Lunatic - Fall Illusion x Lunatic Battle (Top 8)
- + 2018 Lunatic - Fall Performance
- + 2017 D Maniac Megacrew - Hip Hop International Thailand (**Winner**)
- + 2017 D Maniqueen - Hip Hop International Thailand (Ranked 12th)
- + 2016 D Maniqueen - LINE Finance Night Dance Competition (**Winner**)
- + 2016 D Maniqueen - Hip Hop International Thailand (Ranked 6th)
- + 2016-18 KU Dance Club - Kaset Fair Annual Festival
- + 2016 KU Dance Club - Kasetsart University Orientation Day
- + 2015 D Maniqueen - Seacon Street Int'l Challenge (Ranked 6th)
- + 2015 D Maniqueen - To Be Number One Variety Showcase (NBT Channel)
- + 2015 TomYumKoong - Hip Hop International Thailand (Ranked 4th)
- + 2015-16 KU Dance Club - KU First Date
- + 2015-16 KU Dance Club - Freshy Night
- + 2014-16 KU Dance Club - Open Activity World Event
- + 2015-16 KU Dance Club - Royal Project Foundation Annual Event at Central World

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OTHER INTERESTS

Art and Sport

Performing Arts, Acting, Dancing, Singing, Table Tennis, Running, Basketball

Music

K-Ballad, K-Pop, Rap/Hip-Hop, R&B, Soundtracks

Lifestyles

Photography, Dramas/Movies, Mainstream Social Medias